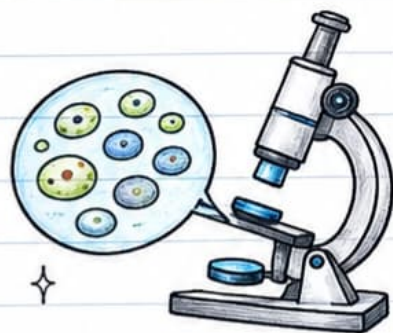


CELL : THE UNIT OF LIFE

Why Cells are called the Unit of Life?

- All living organisms are **made up of cells**.
- Cell is the basic **structural and functional** unit of life.
- All **vital life processes** occur within cells.
- Cells arise only from **pre-existing cells**.



Discovery of Cell

- 1665 - **Robert Hooke** observed cork under a microscope and saw tiny box-like structures which he called **cells**.
- Later - **Anton van Leeuwenhoek** observed living cells in pond water.

Key Point

Cell is the smallest unit of life. ❤️

Types of Cells

1. Prokaryotic Cell -

No true nucleus

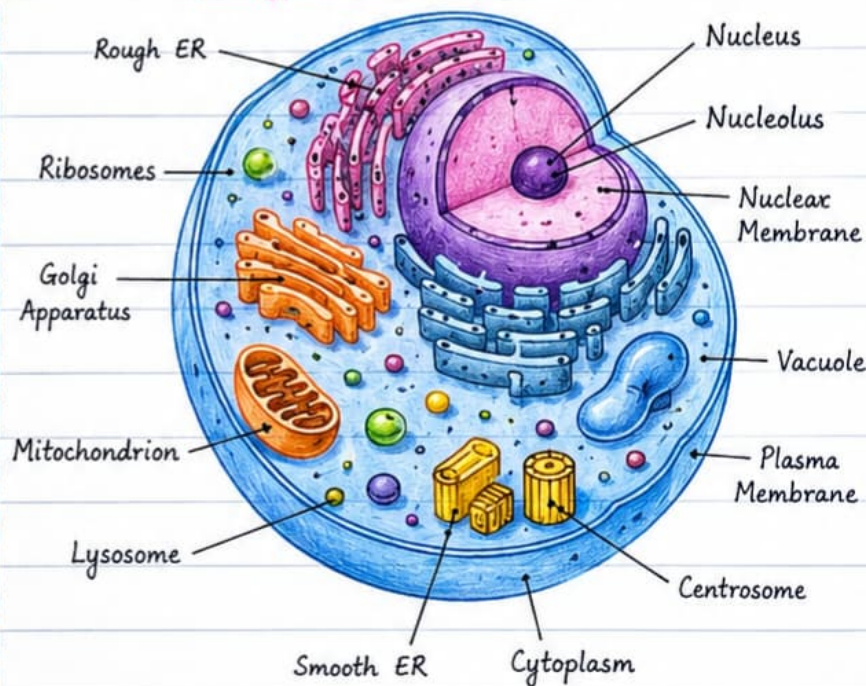
(e.g., Bacteria, Cyanobacteria)

2. Eukaryotic Cell -

True nucleus present

(e.g., Plants, Animals, Fungi)

A Typical Eukaryotic Cell



Basic Components of a Cell

1. Plasma Membrane
2. Cytoplasm
3. Nucleus
4. Organelles
5. Cell Wall (in plants)



Key Takeaway

Cells vary in shape, size and function, but all cells share certain basic components and carry out life processes.



CELL THEORY

- All organisms are made up of **one** or **more** cells.
- Cell is the basic unit of **structure** and **function** in organisms.
- All cells arise from **pre-existing** cells.

Size Range of Cells

- The size of cells varies from **1-2 μm** (bacteria, Mycoplasma) to **100 μm** (bird egg).
- Most plant and animal cells are between **10-30 μm** .

Examples

- ★ **RBC (Human)** - about 7-8 μm
- ★ **Nerve cell** - very long (up to 1 m)
- ★ **Ostrich egg cell** - largest visible cell

Shapes of Cells

- **Spherical** - e.g., RBC
- **Cuboidal** - e.g., Kidney tubule cells
- **Columnar** - e.g., Intestinal cells
- **Spindle shaped** - e.g., Muscle cells
- **Flat / Squamous** - e.g., Cheek cells
- **Branched** - e.g., Nerve cells



Spherical (RBC)



Cuboidal (Kidney cells)



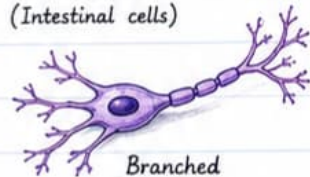
Columnar (Intestinal cells)



Spindle shaped (Muscle cells)



Flat / Squamous (Cheek cells)

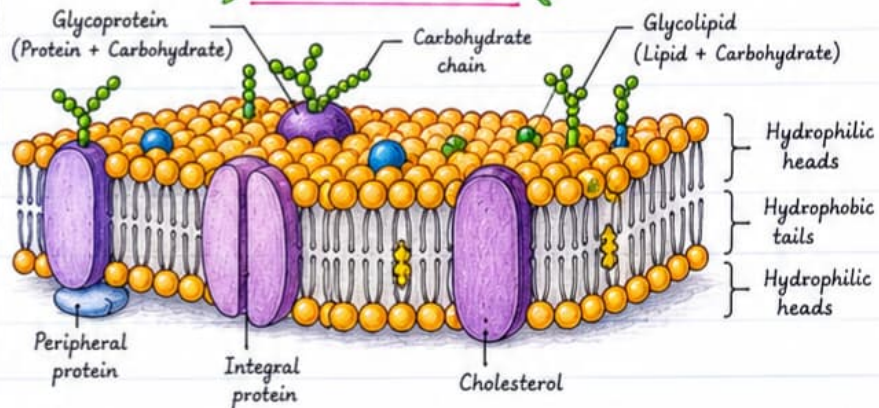


Branched (Nerve cell)

Plasma Membrane

- Also called **cell membrane**.
- It is a thin, living, **selectively permeable** membrane.
- It separates the cell from its surroundings.
- It regulates the movement of substances in **and out** of the cell.

Fluid Mosaic Model

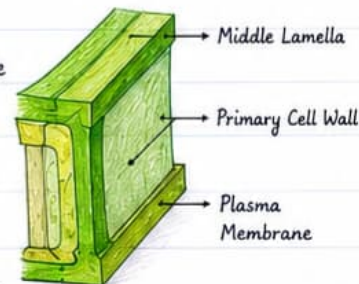


Functions of Plasma Membrane

- ✓ Protects the cell.
- ✓ Maintains cell shape.
- ✓ Regulates entry and exit of substances.
- ✓ Cell recognition and communication.
- ✓ Helps in cell junction and adhesion.

Cell Wall (in Plants)

- Present outside the plasma membrane in **plant cells**.
- Made up of **cellulose**.
- It is non-living, rigid and fully **permeable**.
- Provides **shape, support** and **protection**.

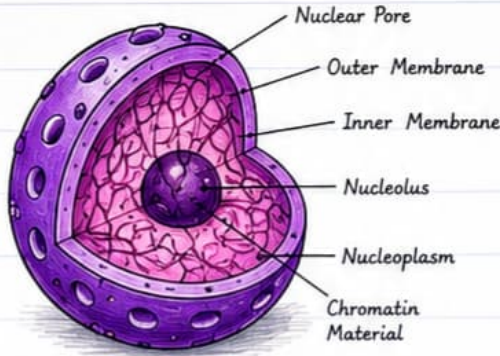


Key Takeaway

Cells are the **basic building blocks** of life. Each cell, enclosed by a plasma membrane (and cell wall in plants), performs various functions essential for life.

Nucleus - The Control Centre

- Usually spherical and bounded by **double** membrane called **nuclear envelope**.
- Encloses **nucleoplasm**, **nucleolus** and **chromatin material**.
- Controls all the **activities** of the cell.
- Contains **genetic material (DNA)** which is responsible for **heredity**.

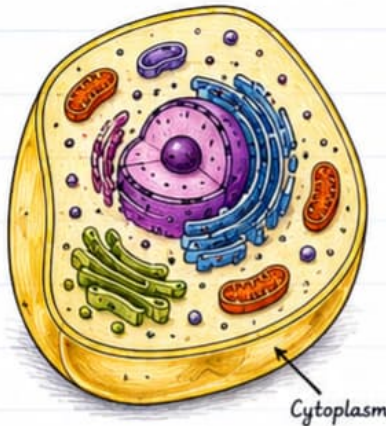


Note

Nucleus is the brain of the cell that stores information and controls functions.

Cytoplasm

- Jelly-like** substance present between plasma membrane and nucleus.
- Contains **organelles**.
- Site of many **metabolic reactions**.
- Helps in the **movement** of cell organelles.



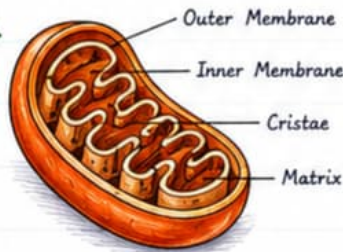
Cell Organelles

Specialised structures in the cytoplasm that perform specific functions.

Organelle	Function
Mitochondria	Powerhouse of the cell; site of respiration.
Endoplasmic Reticulum (ER)	Helps in transport of substances.
Ribosomes	Site of protein synthesis.
Golgi Apparatus	Modifies, packages and secretes materials.
Lysosomes	Contain digestive enzymes; breakdown waste.
Vacuoles	Storage of water, food and wastes.
Centrosome	Helps in cell division.

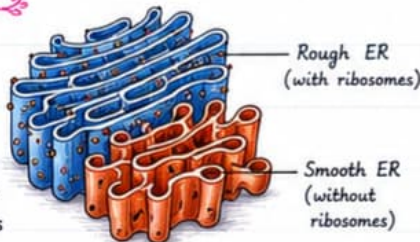
Mitochondria - The Powerhouse

- Double membrane** bound.
- Inner membrane folded into **cristae**.
- Site of **aerobic respiration** and **ATP production**.



Endoplasmic Reticulum (ER)

- Network of **membranous** tubules and sacs.
- Two types :
Rough ER - with ribosomes.
Smooth ER - without ribosomes.
- Helps in transport of materials within the cell.



Ribosomes

- Small, non-membranous particles made of **rRNA** and proteins.
- Site of **protein synthesis**.



Golgi Apparatus

- Made up of flattened membranous sacs called **cisternae**.
- Modifies, sorts and packages materials into vesicles.



Lysosomes

- Small, **spherical** bodies.
- Contain **hydrolytic enzymes**.
- Help in **intracellular digestion** and removal of waste.



Vacuoles

- Membrane bound **sacs**.
- In plants, large central vacuole stores **water**.
- In animals, vacuoles are small and **temporary**.



Centrosome (in animal cells)

- Consists of a pair of **centrioles**.
- Helps in cell division by **spindle formation**.



Summary



- Cell is the basic structural and functional unit of life.
- Cells arise only from pre-existing cells.
- It contains plasma membrane, cytoplasm and nucleus.

- Different organelles perform specific functions.
- Cells are the site of all life processes.

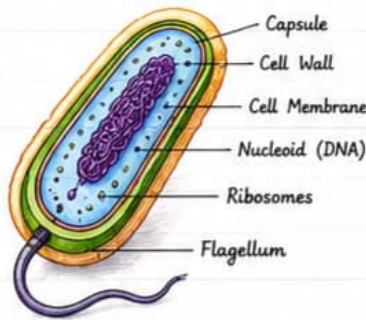


CELL VS NON-LIVING THINGS

Cells (Living)	Non-living Things
• Made up of one or more cells .	• Not made up of cells.
• Show growth .	• Do not grow .
• Carry out metabolism .	• No metabolism .
• Respond to stimuli .	• Do not respond .
• Can reproduce .	• Cannot reproduce .
• Have a limited life span .	• No life span .
• Adapt and evolve .	• Cannot adapt .

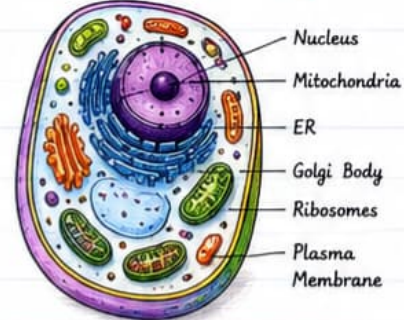
PROKARYOTIC vs EUKARYOTIC CELLS

Prokaryotic Cell (Bacteria)



- No true nucleus (nucleoid).
- No membrane bound organelles.
- Smaller in size (1-10 μm).
- **Example:** Bacteria, Cyanobacteria.

Eukaryotic Cell (Plant/Animal)



- True **nucleus** present.
- Membrane bound organelles present.
- Larger in size (10-100 μm).
- **Example:** Plants, Animals, Fungi.

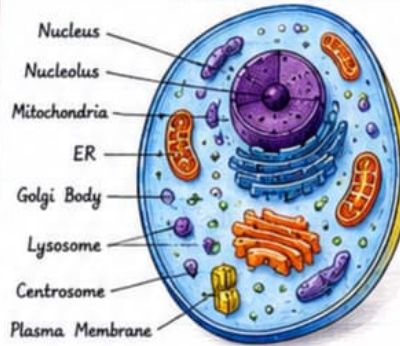
FUNCTIONS OF A CELL

- ✓ Absorbs **nutrients**.
- ✓ Converts nutrients into **energy**.
- ✓ Synthesizes **proteins** and other biomolecules.
- ✓ Removes **waste** products.
- ✓ Responds to internal and external **stimuli**.
- ✓ Helps in **growth** and **repair**.
- ✓ Helps in **reproduction**.



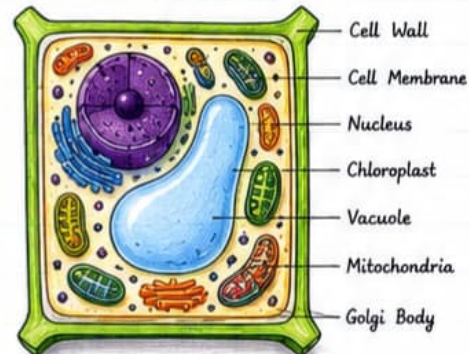
TYPICAL ANIMAL CELL vs TYPICAL PLANT CELL

Animal Cell



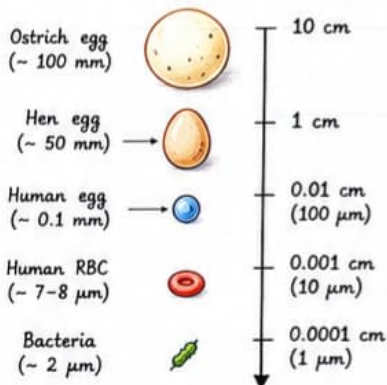
- No cell wall.
- Small vacuoles or absent.
- Centrioles (centrosome) present.
- Shape is **irregular**.

Plant Cell



- Cell wall present (cellulose).
- Large central vacuole.
- No centrioles.
- Shape is fixed (rectangular).

CELL SIZE COMPARISON



1 μm (micrometer) = 10^{-6} meter

MICROSCOPY - Looking into the Microscopic World

1. Light Microscope (LM)



- Uses visible light.
- Magnification: up to 1500 $\times$.
- Used to observe **living cells**.

2. Electron Microscope (EM)



- Uses electron beam.
- Magnification: up to 2,00,000 $\times$.
- Used to observe **ultra-structures**.

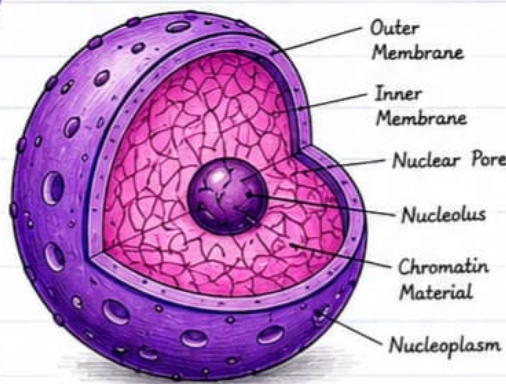
KEY TAKEAWAY

- ★ Cell is the basic structural and functional **unit** of life.
- ★ Cells vary in shape, size and function.
- ★ All cells have **plasma membrane**, **cytoplasm** and genetic material.
- ★ Prokaryotic cells are simple; eukaryotic cells are **complex**.
- ★ Organelles work together to carry out life processes efficiently.



Nucleus - The Control Centre

- The nucleus is the brain of the cell.
- Surrounded by a double membrane called **nuclear envelope**.
- Has **pores** that allow exchange of materials.
- Contains **nucleoplasm**, **nucleolus** and **chromatin material**.
- Nucleolus helps in **ribosome formation**.
- Chromatin contains **DNA**, which carries genetic information.

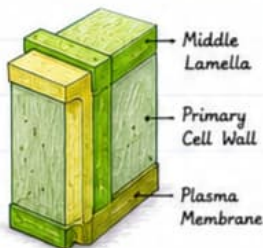


Remember!

Nucleus controls all activities of the cell and passes genetic information from one generation to another.

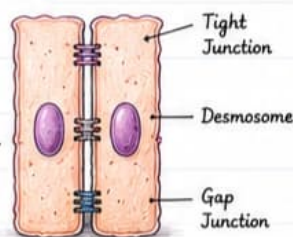
Cell Wall (in Plants)

- Present outside the plasma membrane in **plant cells**.
- Made up of **cellulose**.
- Non-living, rigid and fully **permeable**.
- Provides **shape**, **support** and **protection**.



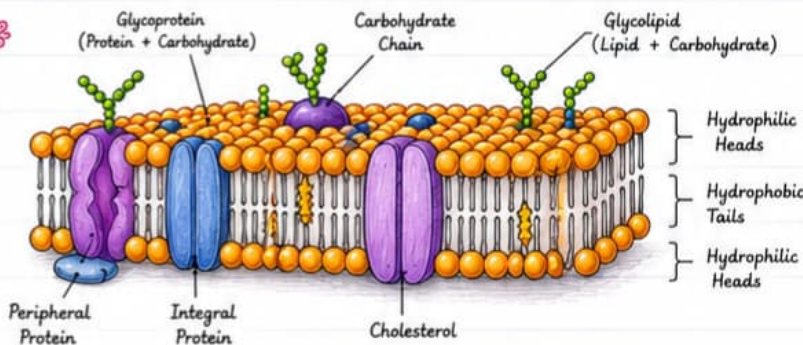
Cell Junctions in Animals

- Cells are held together by specialised junctions.
- Types :
 - ★ **Tight Junctions** - prevent leakage.
 - ★ **Desmosomes** - provide strength.
 - ★ **Gap Junctions** - allow cell to cell communication.



Fluid Mosaic Model of Plasma Membrane

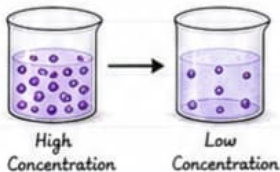
- Proposed by **Singer** and **Nicolson** (1972).
- The membrane is a **dynamic** structure.
- It is a **mosaic** of proteins in a sea of **lipids**.
- Components :
 - Lipids** (phospholipids, glycolipids),
 - Proteins** (integral, peripheral),
 - Carbohydrates** (glycoproteins, glycolipids)



Diffusion and Osmosis

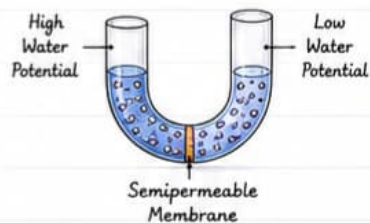
Diffusion

- Movement of particles from **higher** concentration to **lower** concentration.
- No energy required.



Osmosis

- Movement of water from **higher** water potential to **lower** water potential through a **semipermeable** membrane.

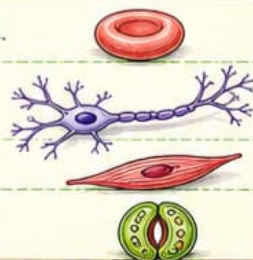


Prokaryotic vs Eukaryotic Cells

Feature	Prokaryotic Cell	Eukaryotic Cell
Size	1 - 10 μm	10 - 100 μm
Nucleus	No true nucleus	True nucleus present
Membrane bound Organelles	Absent	Present
DNA	Single circular DNA	Multiple linear DNA
Examples	Bacteria, Cyanobacteria	Plants, Animals, Fungi

Examples of Cells

- ★ **Human RBC** - Biconcave disc shaped.
- ★ **Nerve Cell** - Long with branched extensions.
- ★ **Muscle Cell** - Long, spindle shaped.
- ★ **Guard Cell (Plant)** - Bean shaped in pairs.



Key Takeaway

- ✓ Cell is the basic structural and functional unit of life.
- ✓ All cells arise from pre-existing cells.
- ✓ Cells contain plasma membrane, cytoplasm, nucleus and organelles.

- ✓ Prokaryotic cells are simple; eukaryotic cells are complex.
- ✓ Cells perform various life processes to sustain life.
- ✓ Understanding cells helps us understand life itself.

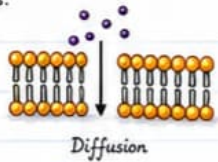
Transport Across Cell Membrane

Plasma membrane is **selectively permeable** and regulates the movement of substances.

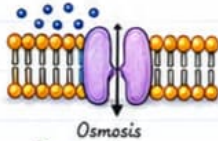
1. Passive Transport (No energy required)

a) **Diffusion** - Movement of molecules from higher to lower concentration.

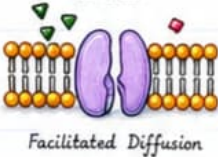
e.g., O_2 , CO_2



b) **Osmosis** - Movement of water from higher to lower water potential.



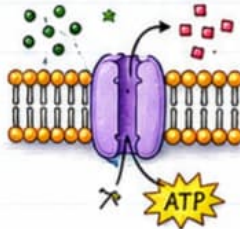
c) **Facilitated Diffusion** - Movement of molecules with the help of carrier proteins.



2. Active Transport (Energy required)

• Movement of molecules from lower to higher concentration with the use of energy (ATP).

e.g., Na^+ , K^+ pump in animal cells.



Differences : Prokaryotic vs Eukaryotic Cells

Feature	Prokaryotic Cell	Eukaryotic Cell
Size	1 - 10 μm	10 - 100 μm
Nucleus	No true nucleus	True nucleus present
Membrane bound Organelles	Absent	Present
DNA	Single circular DNA (Nucleoid region)	Multiple linear DNA (with histone proteins)
Cell Wall	Present (in most) (peptidoglycan)	Present in plants (cellulose), absent in animals
Ribosomes	70S	80S
Cell Division	Binary fission	Mitosis / Meiosis
Examples	Bacteria, Cyanobacteria	Plants, Animals, Fungi

Essence

All life is cellular. Cells are the basic units that build organisms. From tiny bacteria to complex human beings - life begins with a cell.

Examples of Specialised Cells

RBC (Human)



- ✓ Biconcave disc shaped
- ✓ No nucleus
- ✓ Function - Transport of O_2 and CO_2

Nerve Cell



- ✓ Long with branched extensions
- ✓ Function - Transmit nerve impulses

Muscle Cell



- ✓ Long, spindle shaped
- ✓ Function - Contraction and movement

Palisade Cell (Plant)



- ✓ Elongated, closely packed
- ✓ Function - Photosynthesis

Guard Cell (Plant)

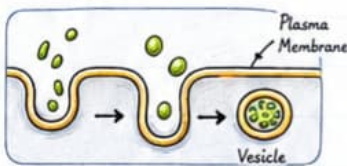


- ✓ Bean shaped (in pairs)
- ✓ Function - Regulates opening & closing of stomata

Endocytosis and Exocytosis

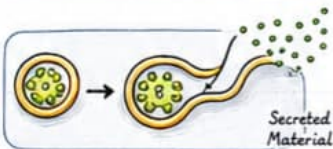
1. Endocytosis (Cell eating)

Cell engulfs solid particles or liquids by forming vesicles. e.g., Amoeba engulfing food.

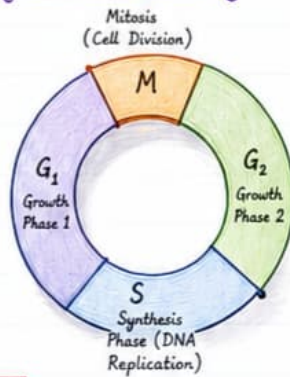


2. Exocytosis (Cell secreting)

Cell releases substances to the outside by fusion of vesicles with the plasma membrane.



Cell Cycle



Stages

- ★ **G₁ Phase** - Cell growth and normal functions.
- ★ **S Phase** - DNA replication.
- ★ **G₂ Phase** - Further growth and preparation for division.
- ★ **M Phase** - Mitosis (nuclear division) and cytokinesis (cytoplasmic division).

Summary

- ✓ Cell is the basic structural and functional unit of life.
- ✓ Cells arise from pre-existing cells.
- ✓ Prokaryotic cells are simpler; eukaryotic cells are complex.
- ✓ Cells contain plasma membrane, cytoplasm, nucleus and organelles.
- ✓ Organelles perform specific functions and work together.
- ✓ Cells vary in shape and size.
- ✓ Cells perform various life processes to sustain life.
- ✓ Understanding cells helps us understand life itself.

